Titanic Dataset – Data Cleaning, Preprocessing & EDA Project

Part 1: Data Loading & Initial Inspection

Question	Task	Action Required
Q1	Load the data. Explore columns data types and report the number of missing values before and after conversion.	Load CSV → Check dtypes → Count missing values → Convert non-numeric age to NaN → Report before/after counts
Q2	Check for exact duplicate rows. Then create a subset of columns `['sex', 'age', 'pclass', 'fare']` and check for duplicates again. If duplicates exist, display them. Should you remove them or fill? Why or why not?	Check exact duplicates → Check subset duplicates → Display if found → Decide remove/fill with justification

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Question	Task	Action Required
Q3	Calculate the median age for each combination of `pclass` and `sex`. Fill missing ages using these medians. Compare original vs imputed age distribution using a histogram.	Groupby (`pclass`, `sex`) → Calculate Part 2: Missing Data Handling median age → Fill NaN → Plot histogram (original vs imputed)

Q4	Find the rows with missing `embarked` values. Look at their `fare` and `pclass`. Find the most common embarkation port for passengers with similar fare and class. Fill the missing values with that port. Explain your reasoning.	Find NaN in `embarked` → Display those rows → Analyze fare/class → Find mode port → Fill → Explain logic
Q5	Calculate the percentage of missing values in `deck`. Create a new column `deck_known` (True if deck known, False if missing). Calculate survival rate for known vs unknown deck. Recommend whether to keep `deck` as a feature or drop it. Justify.	Calculate % missing → Create `deck_known` → Compute survival rate → Compare → Recommend keep/drop with justification

Question	Task	Action Required
Q6	Create `family_size` = `sibsp` + `parch` + 1. Create `is_alone` based on family size. Compare with existing `alone` column. Calculate survival rate by `family_size`. Create a bar plot. Which family size had highest and lowest survival?	Create `family_size` → Create `is_alone` → Compare with `alone` → Groupby survival rate → Bar plot → Identify best/worst size

Question	Task	Action Required
Q7	Analyze survival rate by `who` column. Create `is_child` (age < 18). Compare `who` column's 'child' with `is_child`. Report discrepancies. Who was classified as child by age but not by `who`?	Survival by `who` → Create `is_child` → Compare with `who` → Find mismatches → Report
Q8	Create `fare_per_person = fare / family_size`. Create fare brackets for both `fare` and `fare_per_person`. Calculate survival rate by both. Identify passengers who appear "rich" by total fare but "poor" by per-person fare. Show 3 examples.	Create `fare_per_person` → Create bins → Calculate survival rates → Find mismatched passengers → Show 3 examples
Q9	Group by `sex` and `pclass`. Calculate survival rate and count. Create grouped bar chart. Which class had highest survival for males? Which class had lowest survival for females? Explain.	Groupby (`sex, pclas`[redacted]) → Calculate survival rate & count → Bar chart → Identify highest/lowest → Explain reasons

Q10	Calculate survival rate by `embark_town` alone. Then calculate by `embark_town` AND `pclass`. Calculate class distribution by embarkation port. Is survival difference due to port itself or class composition? Support with numbers.	Survival by town → Survival by town + class → Class distribution by port → Compare → Conclude with numbers
Q11	Group by `alone` and calculate survival rate. Then group by `family_size` and calculate survival rate. Create line plot. Find optimal family size (highest survival) and worst family size (lowest survival). Explain why alone OR large family reduced survival.	Groupby `alone` → Survival rate → Groupby `family_size` → Survival rate → Line plot → Find best/worst → Explain reasons

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Question	Task	Action Required
Q12	Create age bins (0-10, 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80). Calculate survival rate per bin. Create line plot. Create overlayed histogram of survivors vs non-survivors by age. Which age group had highest survival? Which had lowest? Which age group did worse than expected?	Create age bins → Calculate survival per bin → Line plot → Overlayed histogram → Identify highest/lowest → Find unexpected group

Q13	Select only numeric columns. Calculate correlation matrix. Create heatmap. Identify top 3 strongest correlations with `survived`. Find two features highly correlated with each other (not involving survival). What does this tell you?	Select numeric columns → Correlation matrix → Heatmap → Top 3 with `survived` → Find pair correlated → Interpret
Q14	Based on your analysis, identify top 3 factors influencing survival. Create `survival_score` additive score. Find combination with HIGHEST survival rate. Find combination with LOWEST survival rate. Create pivot table of survival rate by top two categorical factors. Write executive summary.	Identify top 3 factors → Create additive score → Find highest combo → Find lowest combo → Pivot table → Write summary